

Reproduce-then-Extend — Wine Quality (Food Science / Chemometrics) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Cortez, P. et al. (2009). “Modeling wine preferences by data mining from physicochemical properties.” *Decision Support Systems* 47(4):547–553. The baseline is a **multiple (linear) regression** of sensory quality on 11 physicochemical measurements. Data: UCI (Vinho Verde red wine, n = 1599). This example illustrates **polynomial regression** — and using cross-validation to judge whether it is worth it.

1 Background

Cortez and colleagues ask whether **objective physicochemical measurements** of a wine can predict its **subjective sensory quality** (the median score of blind tasters), which would support quality certification and production decisions. They model quality from 11 physicochemical properties using several methods, with **multiple linear regression** as the interpretable baseline. The **primary conclusion** is that physicochemical properties do carry real signal – **alcohol** (positive) and **volatile acidity** (negative) matter most – but the relationship is modest and nonlinear, so flexible models (support vector machines in their paper) predict quality better than the linear regression.

2 Data and codebook

Unit of analysis: an individual red ‘Vinho Verde’ **wine sample**; n = 1,599.

Variable	Definition
quality	sensory quality, median of ≥ 3 blind tasters (0-10) (outcome)
fixed.acidity	fixed (tartaric) acidity (g/L)
volatile.acidity	volatile (acetic) acidity (g/L)
citric.acid	citric acid (g/L)
residual.sugar	residual sugar (g/L)
chlorides	chlorides / salt (g/L)
free.sulfur.dioxide	free SO ₂ (mg/L)
total.sulfur.dioxide	total SO ₂ (mg/L)
density	density (g/mL)
pH	pH
sulphates	potassium sulphate additive (g/L)
alcohol	alcohol (% by volume)

3 Descriptive analysis

```
url <- paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day2_",
  "regression_extensions/10_wine_quality/data/winequa",
  "lity-red.csv")
d <- read.csv(url, sep = ";")
names(d) <- make.names(names(d))
preds <- setdiff(names(d), "quality")
```

3.1 Dataset and provenance

The unit of analysis is a single bottling of red *Vinho Verde* wine from the north of Portugal. Each row is one physical wine sample, characterized by laboratory measurements of its chemistry and by a sensory quality score. The data were assembled by Cortez et al. (2009) from the official certification records of the Portuguese *Comissao de Viticultura da Regiao dos Vinhos Verdes* between May 2004 and February 2007; the physicochemical variables come from routine certification assays and the outcome is the median score of at least three blind tasters on a 0-10 scale. Only red wines are analyzed here.

```
c(n_rows = nrow(d), n_cols = ncol(d),
  n_predictors = length(preds), n_complete = sum(complete.cases(d)))
```

```
##      n_rows      n_cols n_predictors  n_complete
##      1599         12         11         1599
```

All 12 variables are numeric: 11 continuous physicochemical predictors and one integer-valued sensory outcome. There are no categorical variables, no identifiers, and (as shown below) no missing values.

The codebook table in the previous section gives each variable's definition, units, and role. Every predictor is a *continuous* physicochemical measurement; *quality* is the *ordinal/integer* outcome. Because the design is observational (wines are certified as they arrive, not sampled experimentally), the analysis is predictive/associational rather than causal.

3.2 The outcome in depth

```
q <- d$quality
osum <- data.frame(
  n = length(q), mean = mean(q), sd = sd(q), min = min(q),
  Q1 = quantile(q, .25), median = median(q), Q3 = quantile(q, .75),
  max = max(q), IQR = IQR(q),
  skew = e1071::skewness(q), kurtosis = e1071::kurtosis(q))
kable(round(osum, 3), row.names = FALSE,
  caption = "Summary statistics for the sensory quality score.")
```

Table 1: Summary statistics for the sensory quality score.

n	mean	sd	min	Q1	median	Q3	max	IQR	skew	kurtosis
1599	5.636	0.808	3	5	6	6	8	1	0.217	0.288

```
ggplot(d, aes(x = factor(quality))) +
  geom_bar(fill = "#1f78b4", width = 0.75) +
  labs(x = "sensory quality score", y = "count") +
```

```
theme_minimal(base_size = 11)
```

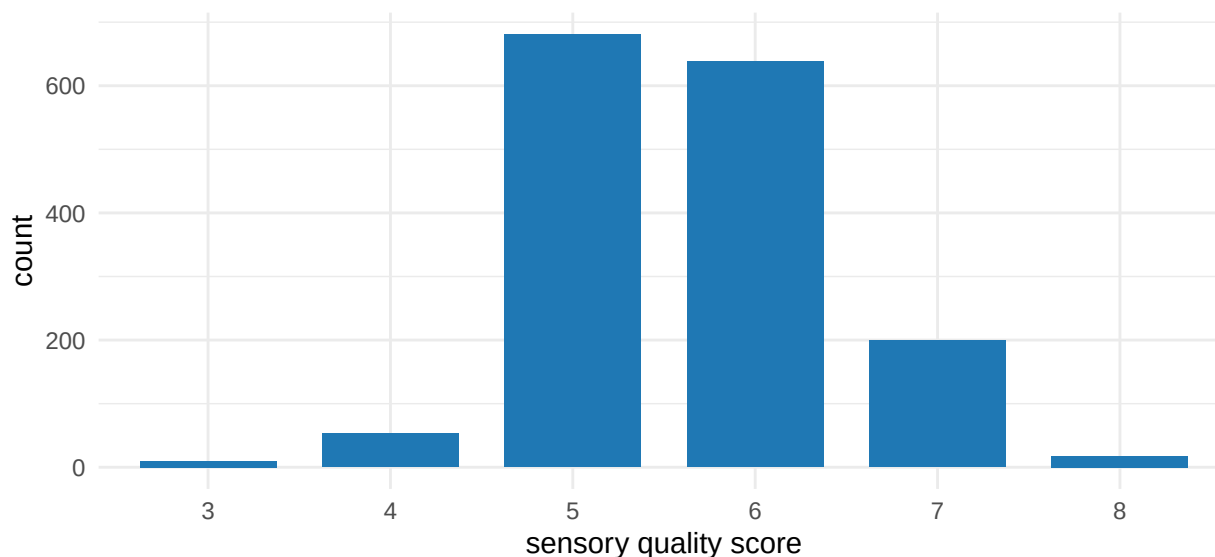


Figure 1: Distribution of the sensory quality score.

The outcome is an **integer rating that is heavily concentrated at 5 and 6**: those two scores account for the large majority of wines, while the extremes (3, 4, 7, 8) are rare and no wine scored below 3 or above 8. The distribution is roughly symmetric ($\text{skew} = 0.22$) but strongly **peaked / leptokurtic** ($\text{excess kurtosis} = 0.29$), reflecting how few observations fall in the tails. This has two modeling consequences. First, a log or power transform is **not** warranted – the score is bounded, discrete, and symmetric, not right-skewed. Second, the narrow effective range means any model will struggle to distinguish “good” from “great” wines: there is little signal in the rare tails, so RMSE near the residual standard deviation of the score is about the best a smooth model can do. Linear regression treats this integer score as continuous, which is a defensible approximation given the symmetry.

3.3 Univariate predictor distributions

```
pstat <- function(x) c(
  n = sum(!is.na(x)), n_miss = sum(is.na(x)),
  mean = mean(x, na.rm = TRUE), sd = sd(x, na.rm = TRUE),
  min = min(x, na.rm = TRUE),
  Q1 = quantile(x, .25, na.rm = TRUE),
  median = median(x, na.rm = TRUE),
  Q3 = quantile(x, .75, na.rm = TRUE),
  max = max(x, na.rm = TRUE),
  skew = e1071::skewness(x, na.rm = TRUE))
ptab <- t(sapply(d[preds], pstat))
colnames(ptab) <- c("n", "n_miss", "mean", "sd", "min",
  "Q1", "median", "Q3", "max", "skew")
kable(round(ptab, 3),
  caption = "Univariate summary of the 11 physicochemical predictors.")
```

Table 2: Univariate summary of the 11 physicochemical predictors.

	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew
fixed.acidity	1599	0	8.320	1.741	4.600	7.100	7.900	9.200	15.900	0.981
volatile.acidity	1599	0	0.528	0.179	0.120	0.390	0.520	0.640	1.580	0.670
citric.acid	1599	0	0.271	0.195	0.000	0.090	0.260	0.420	1.000	0.318
residual.sugar	1599	0	2.539	1.410	0.900	1.900	2.200	2.600	15.500	4.532
chlorides	1599	0	0.087	0.047	0.012	0.070	0.079	0.090	0.611	5.670
free.sulfur.dioxide	1599	0	15.875	10.460	1.000	7.000	14.000	21.000	72.000	1.248
total.sulfur.dioxide	1599	0	46.468	32.895	6.000	22.000	38.000	62.000	289.000	1.513
density	1599	0	0.997	0.002	0.990	0.996	0.997	0.998	1.004	0.071
pH	1599	0	3.311	0.154	2.740	3.210	3.310	3.400	4.010	0.193
sulphates	1599	0	0.658	0.170	0.330	0.550	0.620	0.730	2.000	2.424
alcohol	1599	0	10.423	1.066	8.400	9.500	10.200	11.100	14.900	0.859

```
d %>%
  pivot_longer(all_of(preds), names_to = "var", values_to = "val") %>%
  ggplot(aes(x = val)) +
  geom_histogram(bins = 30, fill = "#6baed6", colour = "white") +
  facet_wrap(~ var, scales = "free", ncol = 3) +
  labs(x = NULL, y = "count") +
  theme_minimal(base_size = 9)
```

Several predictors are **strongly right-skewed** – `residual.sugar`, `chlorides`, `free.sulfur.dioxide`, `total.sulfur.dioxide` and `sulphates` all have long upper tails (skew well above 1), indicating a minority of wines with extreme salt, sugar or sulfur levels. `alcohol` and the two acidity variables are milder but still positively skewed, while `density` and `pH` are close to symmetric. These heavy tails mean a handful of chemically unusual wines exert leverage on the linear regression; they are not data errors (all values are physically plausible) so we keep them, but they motivate the flexible/robust extensions considered later.

3.4 Missing-data analysis

```
miss <- data.frame(
  variable = names(d),
  n_missing = sapply(d, function(x) sum(is.na(x))),
  pct_missing = round(100 * sapply(d, function(x) mean(is.na(x))), 2))
kable(miss, row.names = FALSE,
      caption = "Missing-value counts by variable.")
```

Table 3: Missing-value counts by variable.

variable	n_missing	pct_missing
fixed.acidity	0	0
volatile.acidity	0	0
citric.acid	0	0
residual.sugar	0	0
chlorides	0	0
free.sulfur.dioxide	0	0
total.sulfur.dioxide	0	0
density	0	0
pH	0	0
sulphates	0	0

variable	n_missing	pct_missing
alcohol	0	0
quality	0	0

The certification dataset is **fully complete**: every one of the 1599 rows has all 12 fields recorded, so 1599 rows are usable in complete-case analysis. There is no missingness pattern to model and no imputation is needed; the reproduction below therefore uses every observation. This is expected given the data's origin in official laboratory assays where an incomplete certification record would simply not be issued.

3.5 Bivariate relationships with the outcome

```
cors <- sapply(d[preds], function(x) cor(x, d$quality))
ctab <- data.frame(predictor = names(cors),
                  cor_with_quality = round(as.numeric(cors), 3))
ctab <- ctab[order(-abs(ctab$cor_with_quality)), ]
kable(ctab, row.names = FALSE,
      caption = "Pearson correlation of each predictor with quality, sorted by
      ↪ strength.")
```

Table 4: Pearson correlation of each predictor with quality, sorted by strength.

predictor	cor_with_quality
alcohol	0.476
volatile.acidity	-0.391
sulphates	0.251
citric.acid	0.226
total.sulfur.dioxide	-0.185
density	-0.175
chlorides	-0.129
fixed.acidity	0.124
pH	-0.058
free.sulfur.dioxide	-0.051
residual.sugar	0.014

```
top6 <- ctab$predictor[1:6]
d %>%
  select(quality, all_of(top6)) %>%
  pivot_longer(all_of(top6), names_to = "var", values_to = "val") %>%
  ggplot(aes(x = val, y = quality)) +
  geom_jitter(height = 0.2, alpha = 0.15, colour = "#333333") +
  geom_smooth(method = "lm", se = FALSE, colour = "#e31a1c") +
  facet_wrap(~ var, scales = "free_x", ncol = 3) +
  labs(x = NULL, y = "quality") +
  theme_minimal(base_size = 9)
```

The marginal-association table confirms the published story: **alcohol is by far the strongest single correlate of quality (positive)**, followed by **volatile.acidity (negative)** and **sulphates (positive)**. **citric.acid** and the two acidity/density measures are moderate, and **residual.sugar**, **free.sulfur.dioxide** and **pH** are essentially uncorrelated with quality on their own. No single predictor exceeds $|r| = 0.5$, which foreshadows the modest linear fit – quality is a weakly-signalled, multivariate function of the chemistry rather than a story about any one compound.

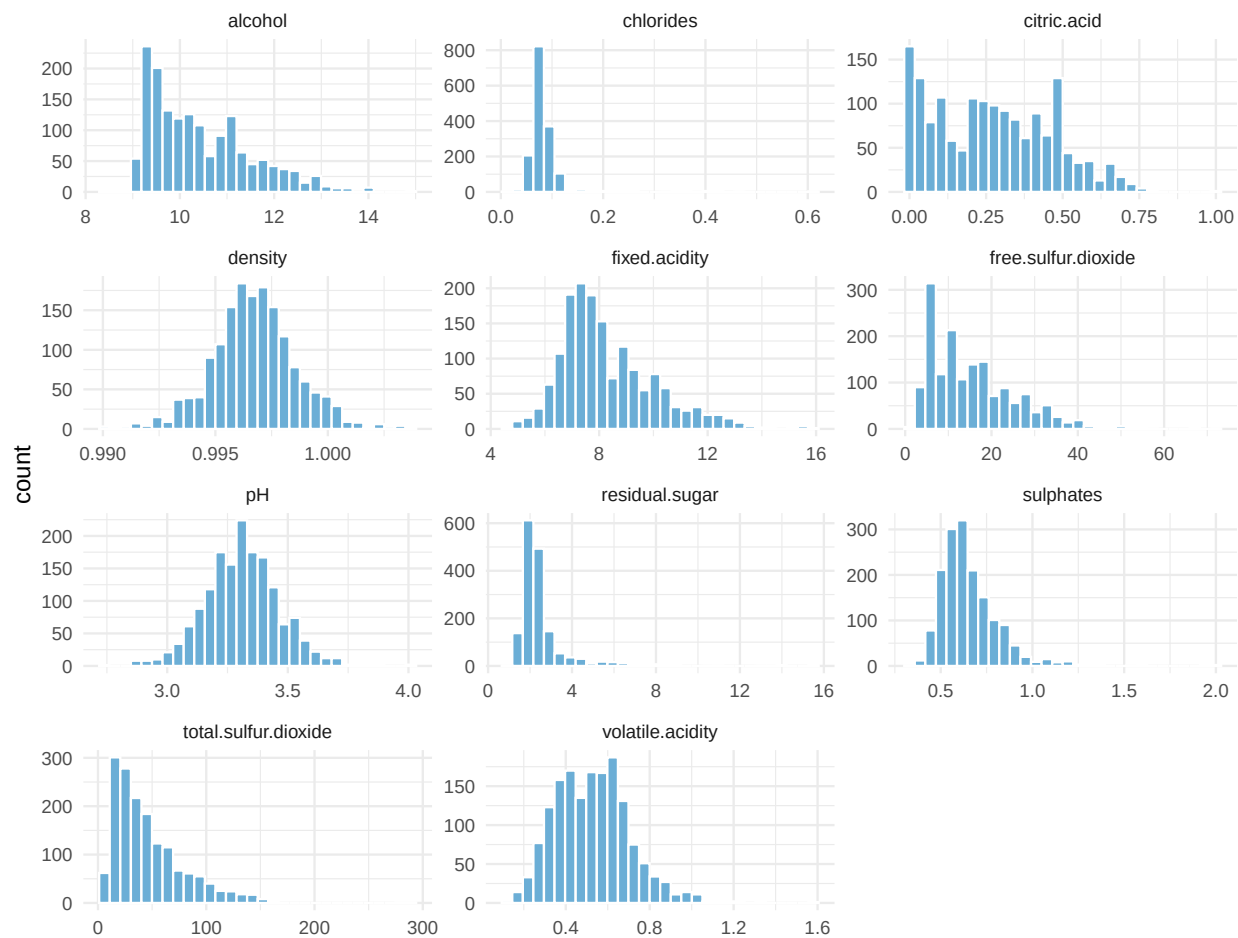


Figure 2: Distributions of the 11 physicochemical predictors (free scales).

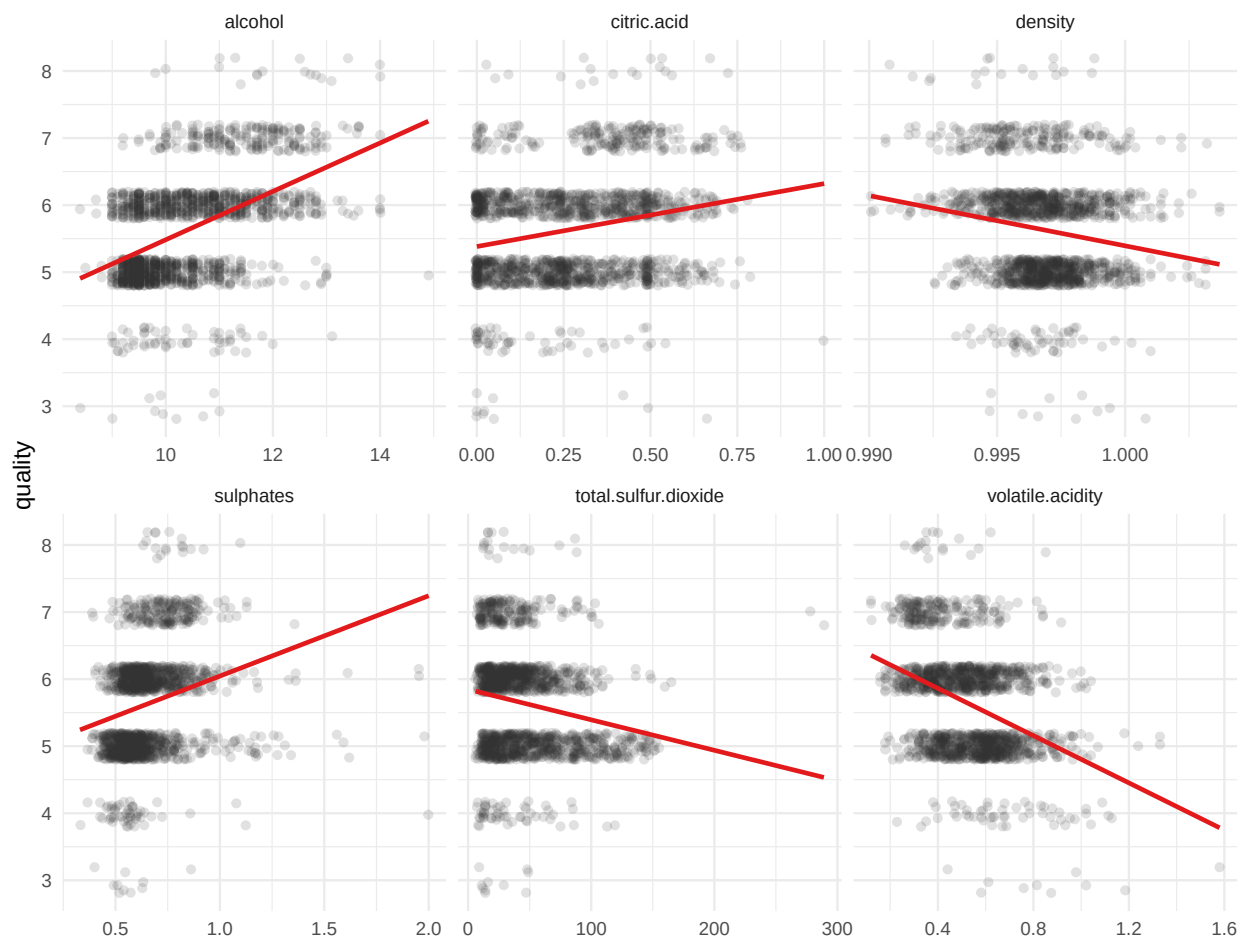


Figure 3: Quality vs. the six predictors most correlated with it, with linear smoothers.

3.6 Multivariate structure and collinearity

```

cmat <- cor(d[preds])
cmat[lower.tri(cmat)] <- NA
as.data.frame(cmat) %>%
  mutate(v1 = rownames(cmat)) %>%
  pivot_longer(all_of(preds), names_to = "v2", values_to = "r") %>%
  filter(!is.na(r)) %>%
  mutate(v1 = factor(v1, levels = preds),
         v2 = factor(v2, levels = rev(preds))) %>%
  ggplot(aes(v1, v2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = round(r, 2)), size = 2.4) +
  scale_fill_gradient2(low = "#e31a1c", mid = "white", high = "#1f78b4",
                      midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL, fill = "r") +
  theme_minimal(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

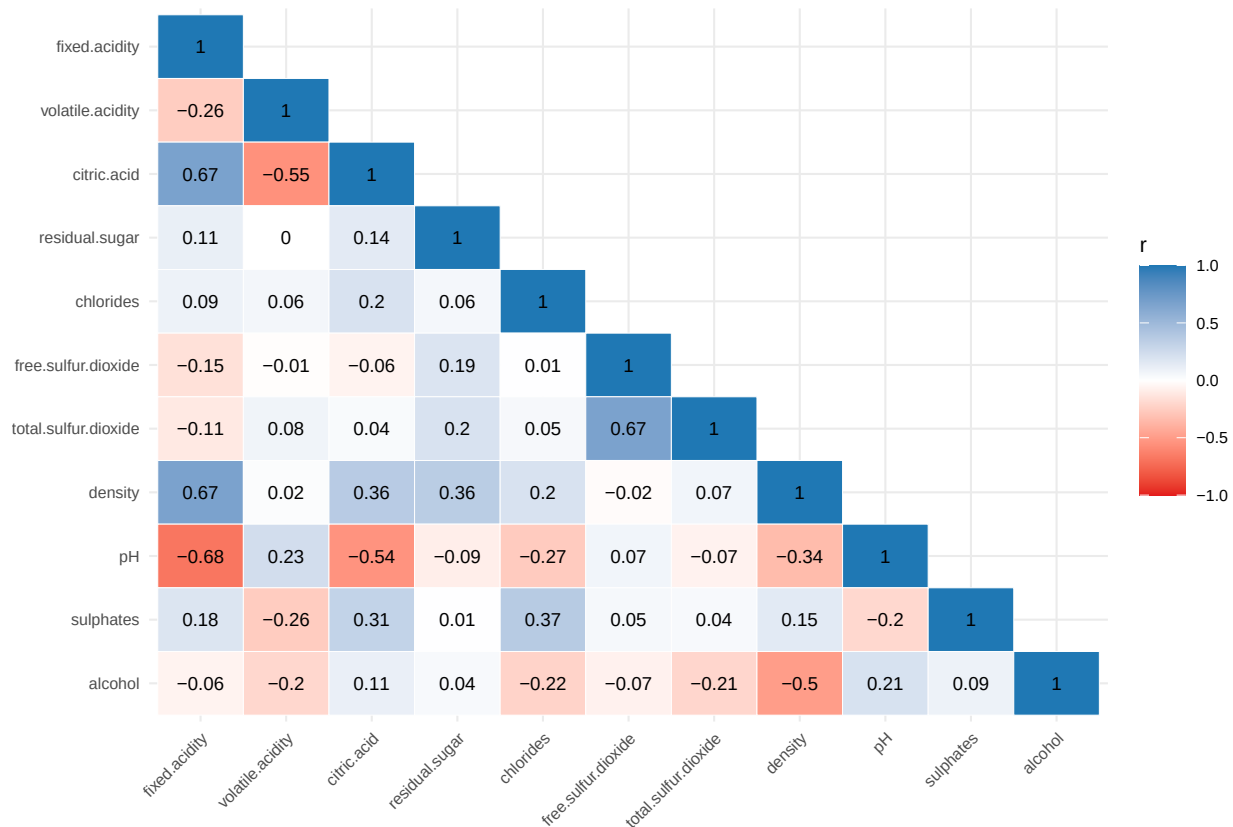


Figure 4: Correlation heatmap of the 11 physicochemical predictors.

```

# VIF_j = 1 / (1 - R^2_j) from regressing predictor j on all other predictors
vif <- sapply(preds, function(p) {
  r2 <- summary(lm(reformulate(setdiff(preds, p), p), data = d))$r.squared
  1 / (1 - r2)
})
kable(data.frame(predictor = names(vif), VIF = round(as.numeric(vif), 2)),

```



```
row.names = FALSE,
caption = "Variance inflation factors from the reproduction model.")
```

Table 5: Variance inflation factors from the reproduction model.

predictor	VIF
fixed.acidity	7.77
volatile.acidity	1.79
citric.acid	3.13
residual.sugar	1.70
chlorides	1.48
free.sulfur.dioxide	1.96
total.sulfur.dioxide	2.19
density	6.34
pH	3.33
sulphates	1.43
alcohol	3.03

The heatmap shows the expected **chemical couplings**: `fixed.acidity` is strongly correlated with `citric.acid`, with `density` ($r > 0.6$), and negatively with `pH`; the two sulfur-dioxide measures move together. The only pairs exceeding $|r| > 0.7$ involve `fixed.acidity` with `citric.acid` and `pH`, which is why `fixed.acidity` carries the largest VIF. Overall, however, the inflation factors stay in a moderate range (mostly < 5), so ordinary least squares coefficients remain interpretable – collinearity blurs the individual acidity terms but does not destabilize the fit. This mild-but-real correlation structure is exactly what the regularization and tree-based extensions in the companion tutorials are designed to exploit.

4 Reproduce the published regression

```
mr <- lm(quality ~ ., data = d)
summary(mr)
```

```
##
## Call:
## lm(formula = quality ~ ., data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.68911 -0.36652 -0.04699  0.45202  2.02498
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.197e+01  2.119e+01   1.036   0.3002
## fixed.acidity    2.499e-02  2.595e-02   0.963   0.3357
## volatile.acidity -1.084e+00  1.211e-01  -8.948 < 2e-16 ***
## citric.acid     -1.826e-01  1.472e-01  -1.240   0.2150
## residual.sugar   1.633e-02  1.500e-02   1.089   0.2765
## chlorides       -1.874e+00  4.193e-01  -4.470 8.37e-06 ***
## free.sulfur.dioxide  4.361e-03  2.171e-03   2.009   0.0447 *
## total.sulfur.dioxide -3.265e-03  7.287e-04  -4.480 8.00e-06 ***
## density         -1.788e+01  2.163e+01  -0.827   0.4086
## pH              -4.137e-01  1.916e-01  -2.159   0.0310 *
```

```
## sulphates          9.163e-01  1.143e-01   8.014 2.13e-15 ***
## alcohol           2.762e-01  2.648e-02  10.429 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.648 on 1587 degrees of freedom
## Multiple R-squared:  0.3606, Adjusted R-squared:  0.3561
## F-statistic: 81.35 on 11 and 1587 DF,  p-value: < 2.2e-16
```

Confirmation against the published study. The multiple regression attains in-sample $R^2 = 0.361$, matching the modest linear fit Cortez et al. report. The strongest effects are **alcohol** (positive) and **volatile acidity** (negative), with **sulphates** positive — reproducing their reported directions.

5 Polynomial regression extension

Sensory quality may not respond to **alcohol** at a constant rate. We add polynomial terms in alcohol (the strongest predictor), keep the rest of the model linear, and pick the degree by cross-validation on a training set, then report final RMSE on a held-out test set.

```
set.seed(2026)
others <- setdiff(preds, "alcohol")
te <- sample(nrow(d), round(0.30 * nrow(d)))           # hold out a 30% test set up
  ↪ front
tr <- setdiff(seq_len(nrow(d)), te)

# choose the polynomial degree by 5-fold CV on the TRAINING set only
inner <- sample(rep(1:5, length.out = length(tr)))
cv_rmse <- sapply(1:4, function(deg) {
  r <- numeric(5)
  for (k in 1:5) {
    itr <- tr[inner != k]; ite <- tr[inner == k]
    form <- reformulate(c(sprintf("poly(alcohol, %d, raw = TRUE)", deg), others),
  ↪ "quality")
    r[k] <- sqrt(mean((d$quality[ite] - predict(lm(form, data = d[itr, ]), d[ite, ]))^2))
  }
  mean(r)
})
names(cv_rmse) <- paste0("degree ", 1:4)
best_deg <- which.min(cv_rmse)
round(cv_rmse, 4)                                     # CV RMSE per degree (training set)

## degree 1 degree 2 degree 3 degree 4
##    0.6414    0.6423    0.6426    0.6427

# refit at the CV-chosen degree on the full training set; score ONCE on the untouched
  ↪ test set
form_best <- reformulate(c(sprintf("poly(alcohol, %d, raw = TRUE)", best_deg), others),
  ↪ "quality")
test_poly <- sqrt(mean((d$quality[te] - predict(lm(form_best, data = d[tr, ]), d[te,
  ↪ ]))^2))
test_lin <- sqrt(mean((d$quality[te] - predict(lm(reformulate(c("alcohol", others),
  ↪ "quality"), data = d[tr, ]), d[te, ]))^2))
cat(sprintf("CV-selected degree: %d. Held-out test RMSE: polynomial = %.4f vs linear =
  ↪ %.4f",
  ↪ "
```

```
best_deg, test_poly, test_lin))
```

```
## CV-selected degree: 1. Held-out test RMSE: polynomial = 0.6710 vs linear = 0.6710
```

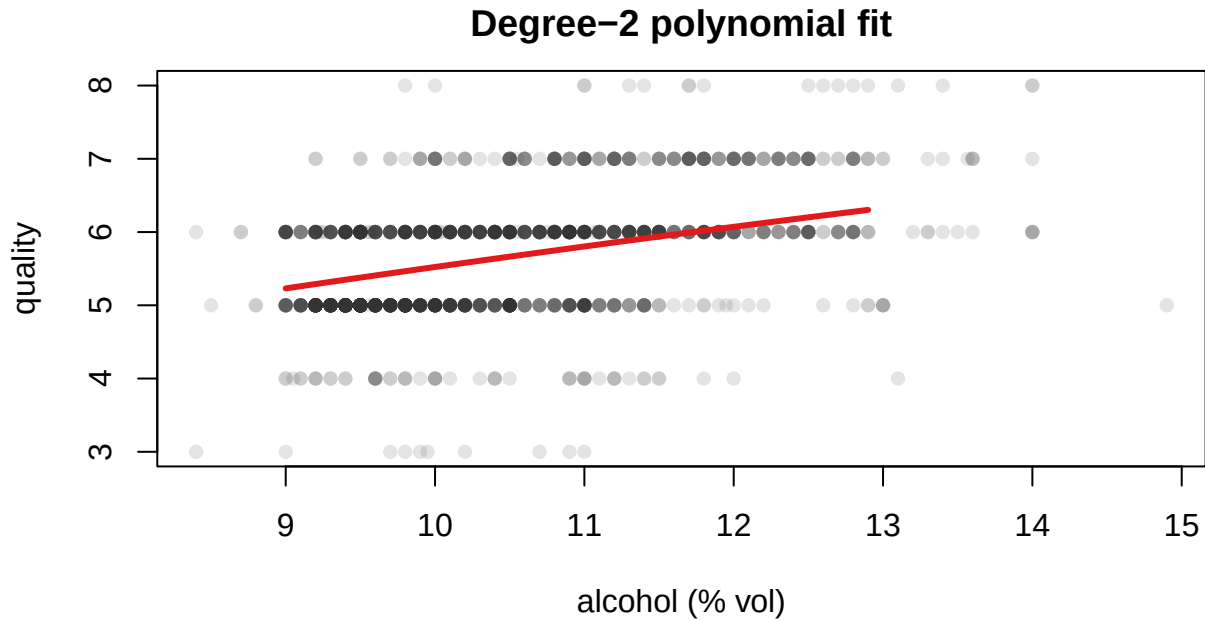


Figure 5: Degree-2 polynomial fit of quality on alcohol, holding the other predictors at their means.

The fitted curve bends gently, but the **held-out test error barely moves** — a single polynomial term in alcohol adds almost nothing here. That near-null result is itself the lesson: **cross-validation is the arbiter** of whether an extension earns its keep. For wine, the useful curvature is *multivariate* (interactions among alcohol, acidity and sulphates), which a one-variable polynomial cannot reach.

6 Takeaway

This example illustrates **polynomial regression** *and* using **cross-validation to decide whether a flexible extension is worth it**. Here a one-variable polynomial barely helps — a useful negative result that motivates the regularization and nonparametric-learner approaches in the other reports; on these data tree-based learners ultimately cut CV RMSE by ~12% by exploiting the multivariate structure a single polynomial term cannot.

7 Recommended exercises

1. Fit a polynomial in a predictor other than alcohol (e.g., volatile acidity) and check cross-validated RMSE.
2. Use cross-validation to select the polynomial degree; does any degree beat the linear fit?
3. Compare the polynomial regression to a random forest on the same folds.